

Self-Learning: 80386 Memory Management & MESI Cache Protocol -- MPCA Module 5

Definition: The 80386 Memory Management Unit (MMU) provides two-stage address translation (Segmentation & Paging), while multiprocessor systems maintain cache consistency using the MESI (Illinois) Protocol.

1. 80386 Protected Mode Memory Translation

```

Logical Address (Selector:Offset)
    |
    v
[ SEGMENTATION UNIT ] ---> Checks Descriptor in GDT/LDT -> Produces 32-bit Linear Address
    |
    v
[ PAGING UNIT (PG=1)] ---> Uses CR3, Page Directory & Page Table -> Produces 32-bit Physical Address
  
```

1. Descriptor Tables (GDT, LDT, IDT):

- * GDT (Global Descriptor Table): Holds segment descriptors accessible by all system tasks.

2. Two-Level Paging Scheme (4 KB Pages):

A 32-bit Linear Address is split into:

- * CR3 holds base address of Page Directory.

2. MESI Cache Coherence Protocol

In symmetric multiprocessing (SMP) systems with private L1/L2 caches, every cache line exists in one of four MESI states:

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M - MODIFIED : Cache line is dirty (present only in current cache, modified, memory is stale).
E - EXCLUSIVE : Cache line is clean (present only in current cache, matches main memory).
S - SHARED   : Cache line is clean (present in multiple processor caches, matches main memory).
I - INVALID  : Cache line is invalid (contains obsolete/stale data; must fetch from bus).
  
```

State Transition Summary:

- * Read Hit: Stays in M, E, or S state.

3. Quick Recall Flow

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Logical (Selector:Offset) -> GDT/LDT -> Linear Address -> Page Dir -> Page Table -> Physical Address | MESI (Modified, Exclusive, Shared, Invalid)
  
```